

RAPID PROTOTYPING MOBILE PROJECTS WITH ARDUINO AND OPEN HARDWARE

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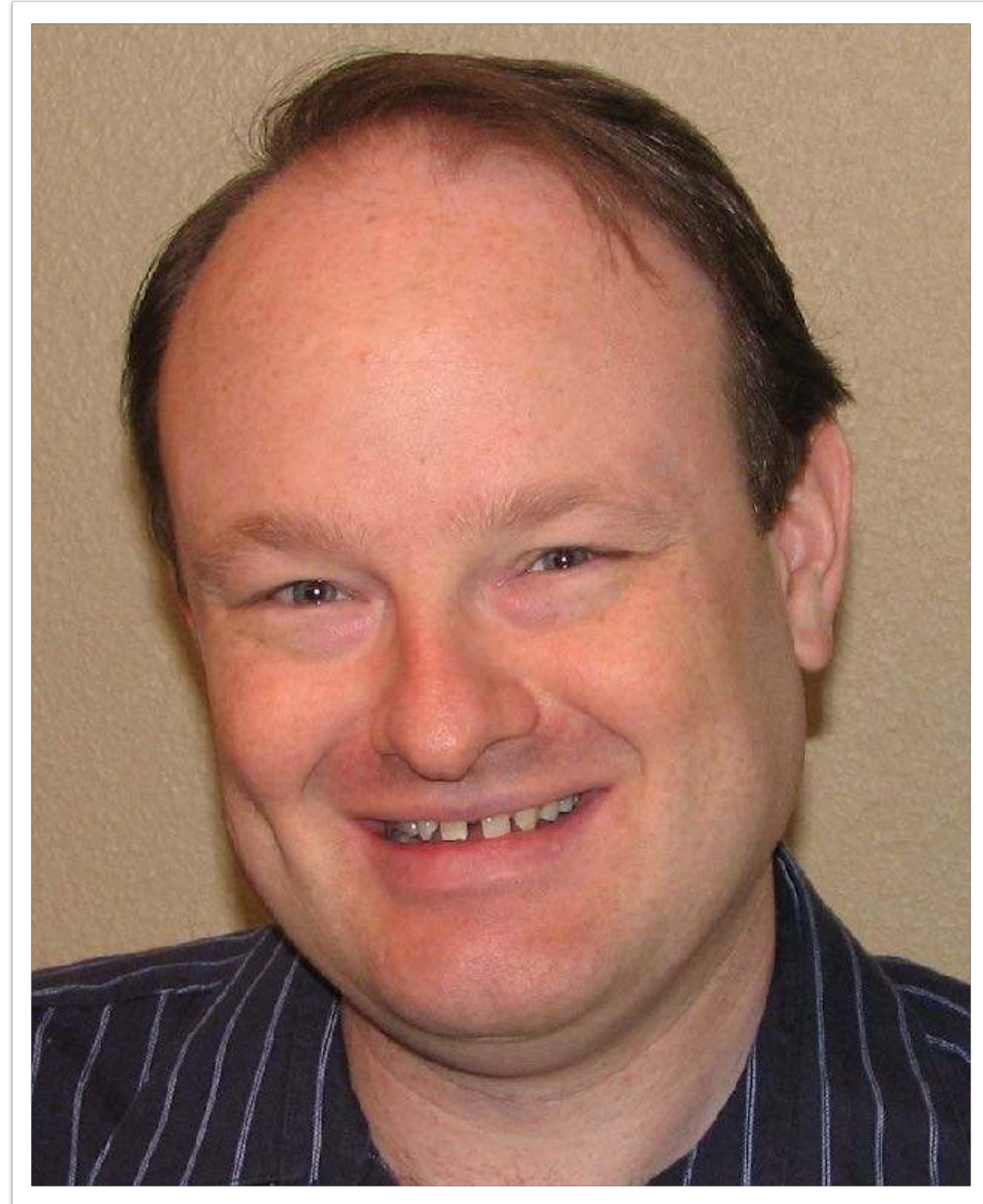


ACKNOWLEDGEMENT & DISCLAIMER

Part of my talk covers products that are for sale by different companies. Some by my employer. I am not a sales person, and am not on any sort of commission. I'm here sharing stuff I'm excited about. I will point out free options and discuss the differences too.

ABOUT YOUR PRESENTER - JIM McKEETH

- Chief Developer Advocate & Engineer
 - Long time software developer
- Invented and patented pattern and swipe to unlock
 - US Patent # 8352745 & 6766456, etc.
- Built thought controlled drone with Google Glass
- Host of Podcast at Delphi.org
- Lives near Boise, Idaho, USA with family & dogs
- Improvisational comedy performer with CSz Boise
- Contributing author to *Internet of Things and Big Data Handbook*



ABOUT EMBARCADERO TECHNOLOGIES

- Elite Developer tools
- Tools used by 90% of the Fortune 100 companies
- Over 30 years of award winning products
- Emphasis on performance, productivity and platforms
- Active developer community of over 1 million developers
- Products include:
 - RAD Studio, RAD Server, Delphi, C++Builder, RAD Server, BeaconFence & InterBase
- Supported platforms: Windows, Android, iOS, macOS & Linux



A photograph of a city skyline at sunset. The sun is low on the horizon, creating a bright orange and yellow glow. The sky is filled with wispy clouds. The city buildings are silhouetted against the bright sky. The quote is overlaid on the lower part of the image.

"THE BEST WAY TO PREDICT THE FUTURE IS TO INVENT IT."

- ALANKAY



SESSION GOAL

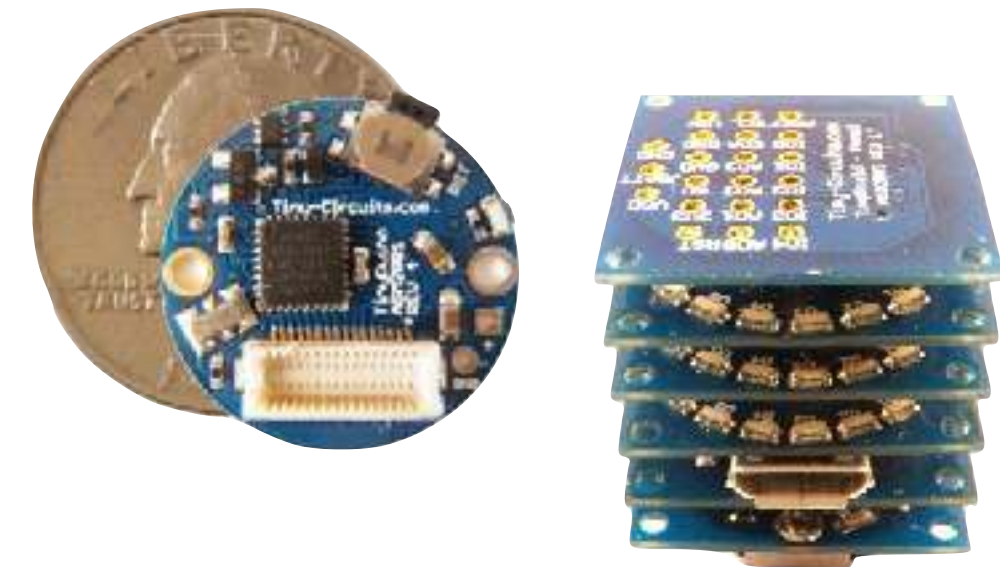
Overview of tools to get from idea
to prototype *quickly*

AGENDA

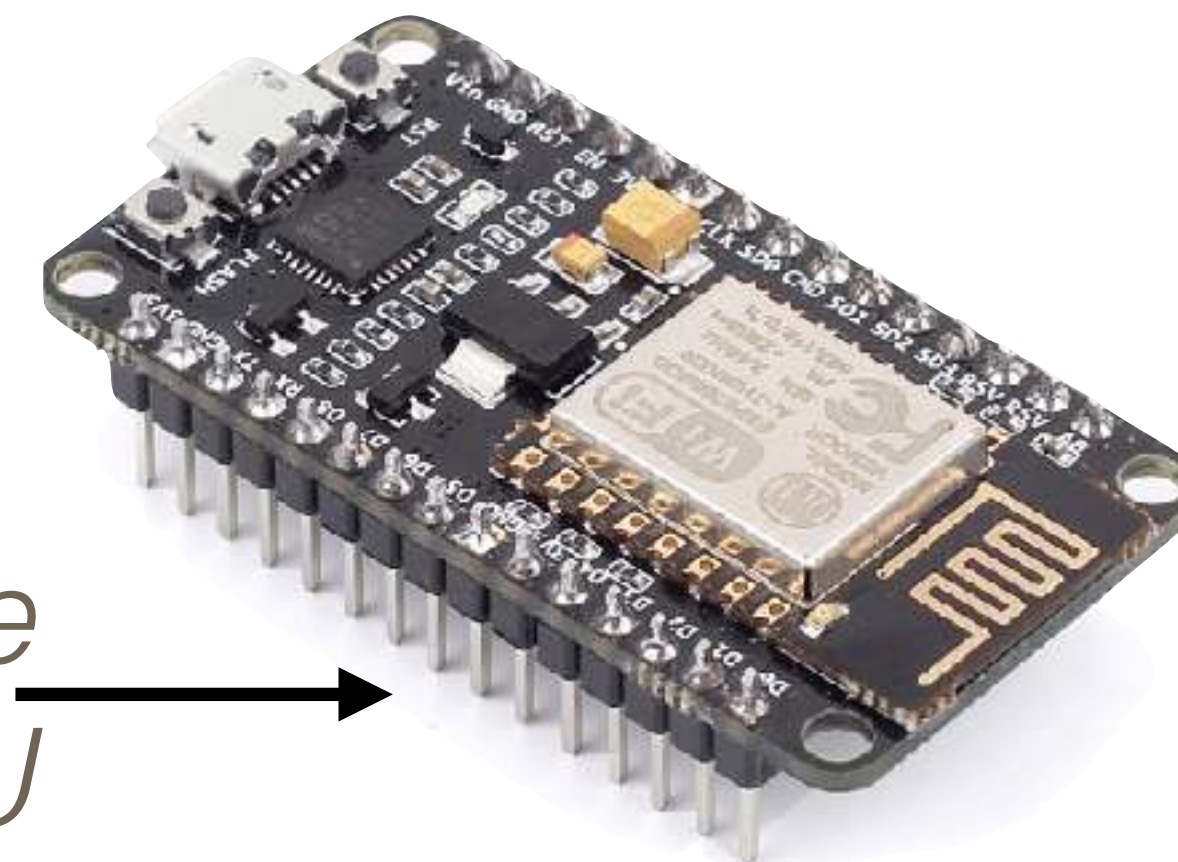
- Introduction to Arduino & development
- Looking at a *few* different hardware pieces
- Overview of Visuino - Visual Arduino development IDE
- Arduino connection options
- Mobile app development
- Architectural considerations
- Cloud options

WHAT IS ARDUINO

- A Company & A Brand
- Open source software & hardware (GPL, LGPL & CC)
- A Maker Community
- There are *compatible* & *derived* boards
- Beware the *clones*



Compatible
NodeMCU



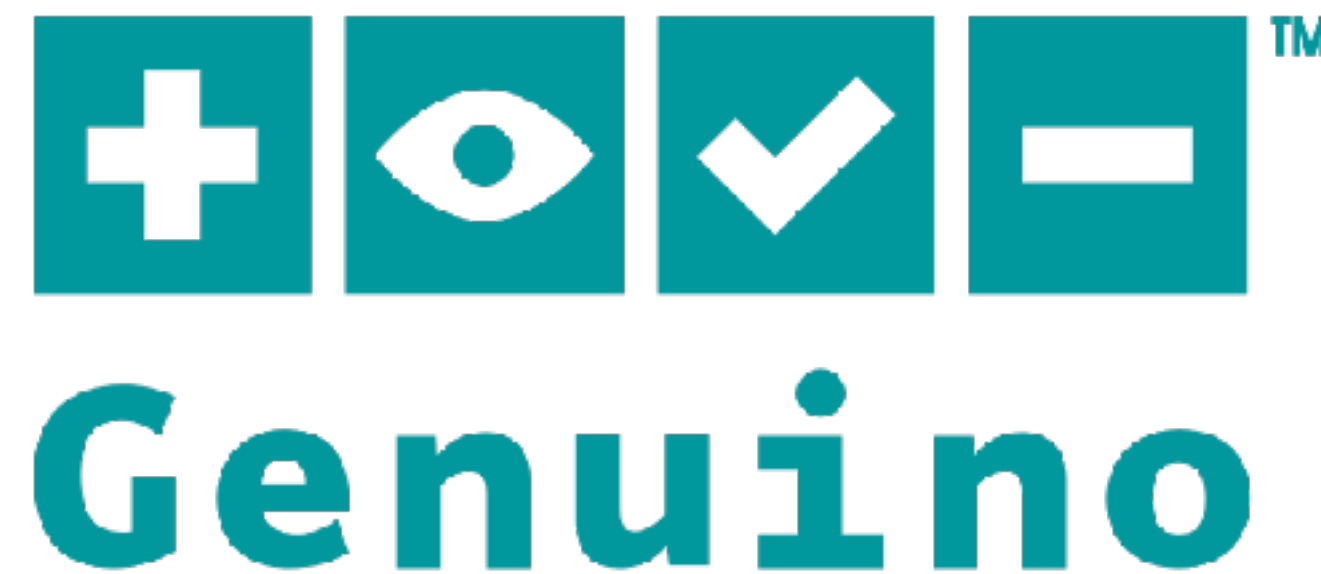
THE DIFFERENT FACES OF ARDUINO

The Company
www.arduino.cc



Open Source
Platform / Ecosystem
www.arduino.org

Brand Outside USA



Built w/Arduino
Compatible
Microprocessors



Community Groups
(Not hardware)



Manufacturing Partners

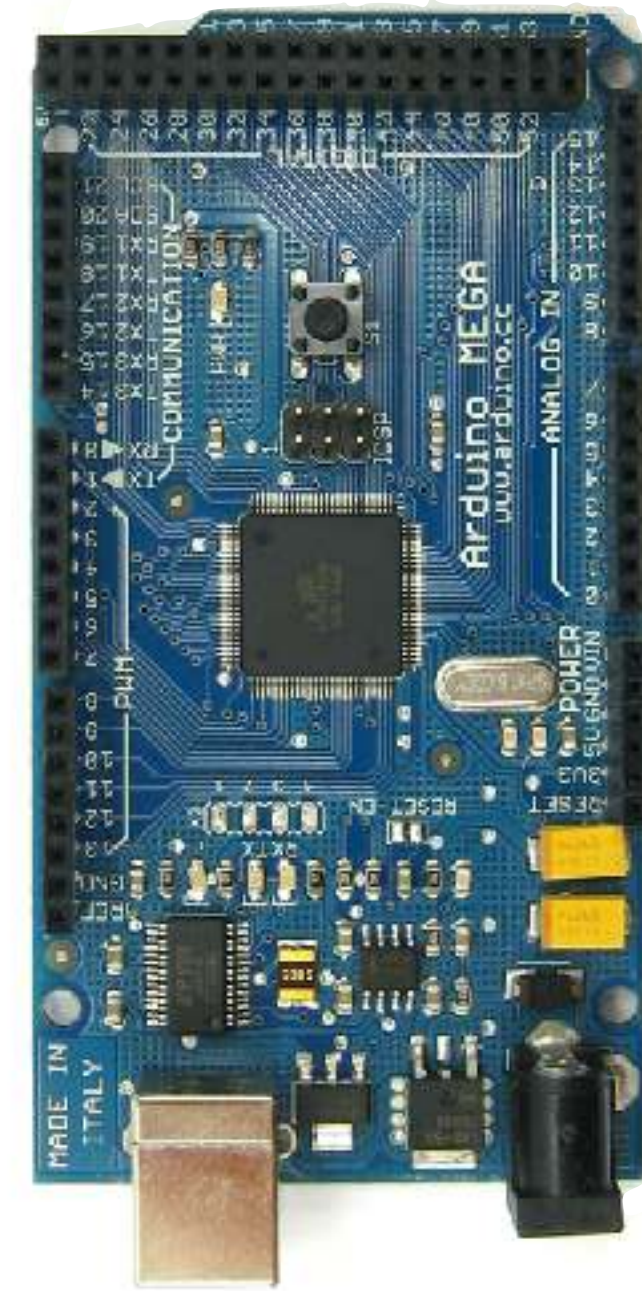
SOME OFFICIAL ARDUINO BOARDS



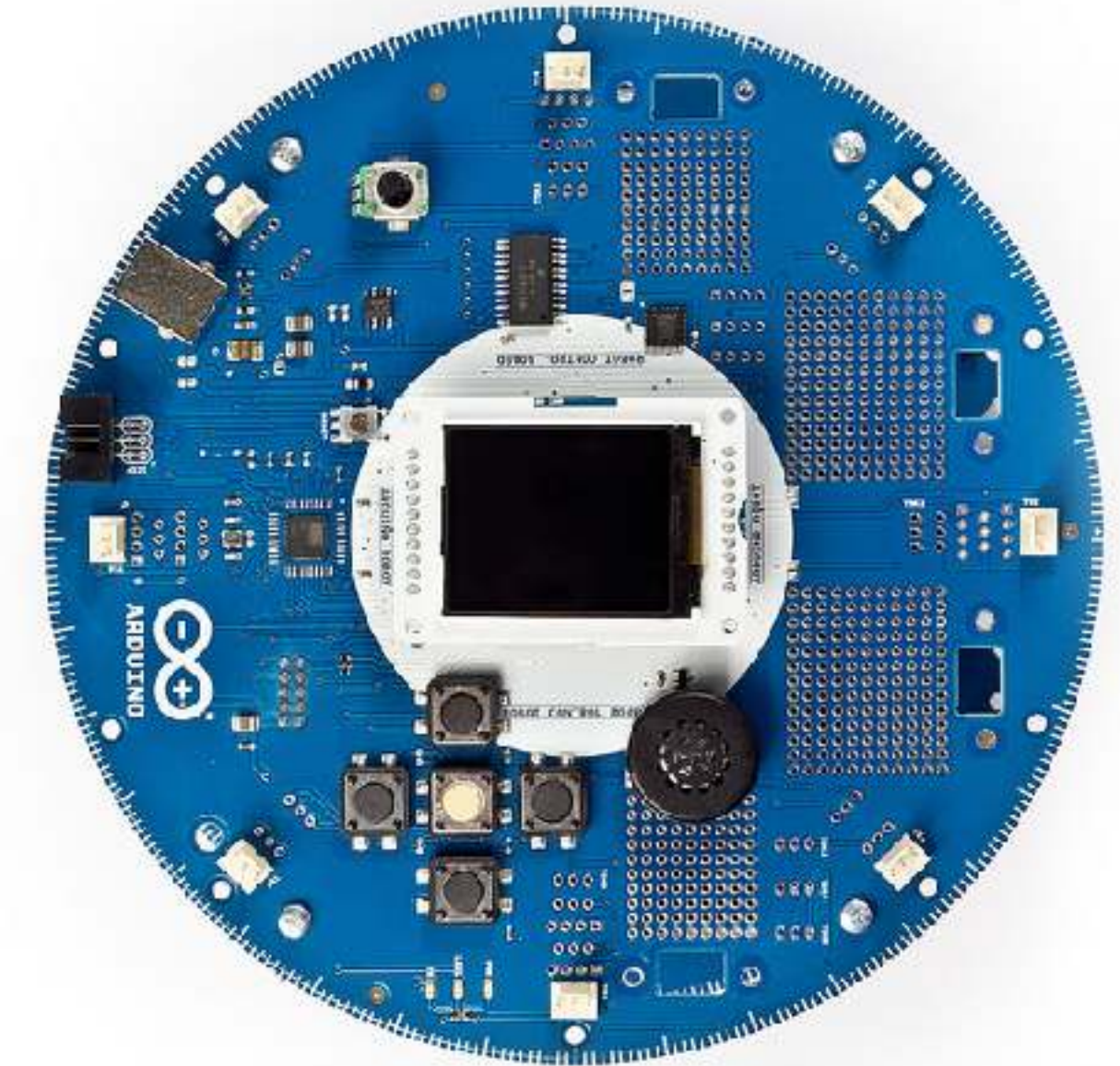
Nano



Uno

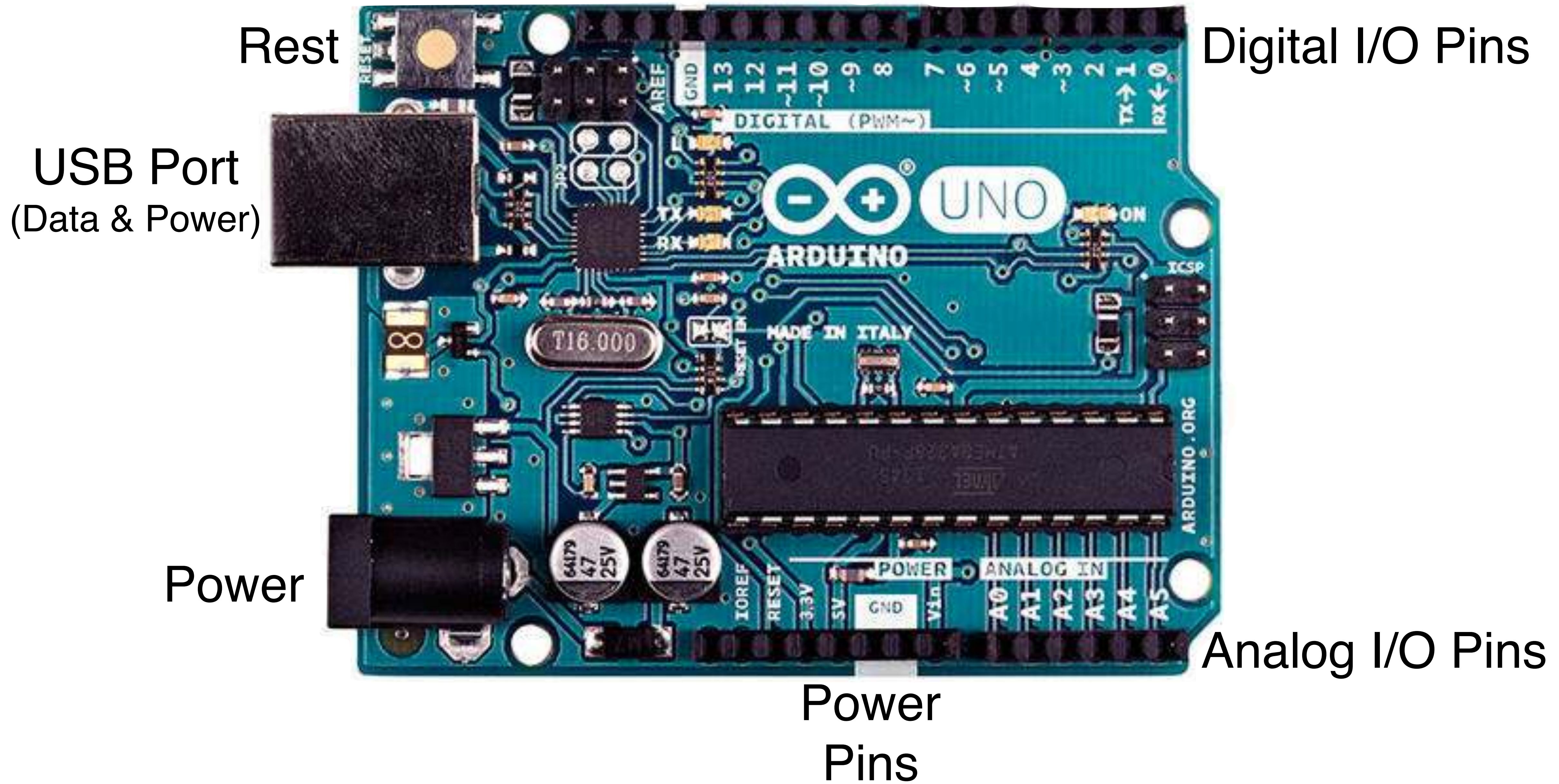


Mega

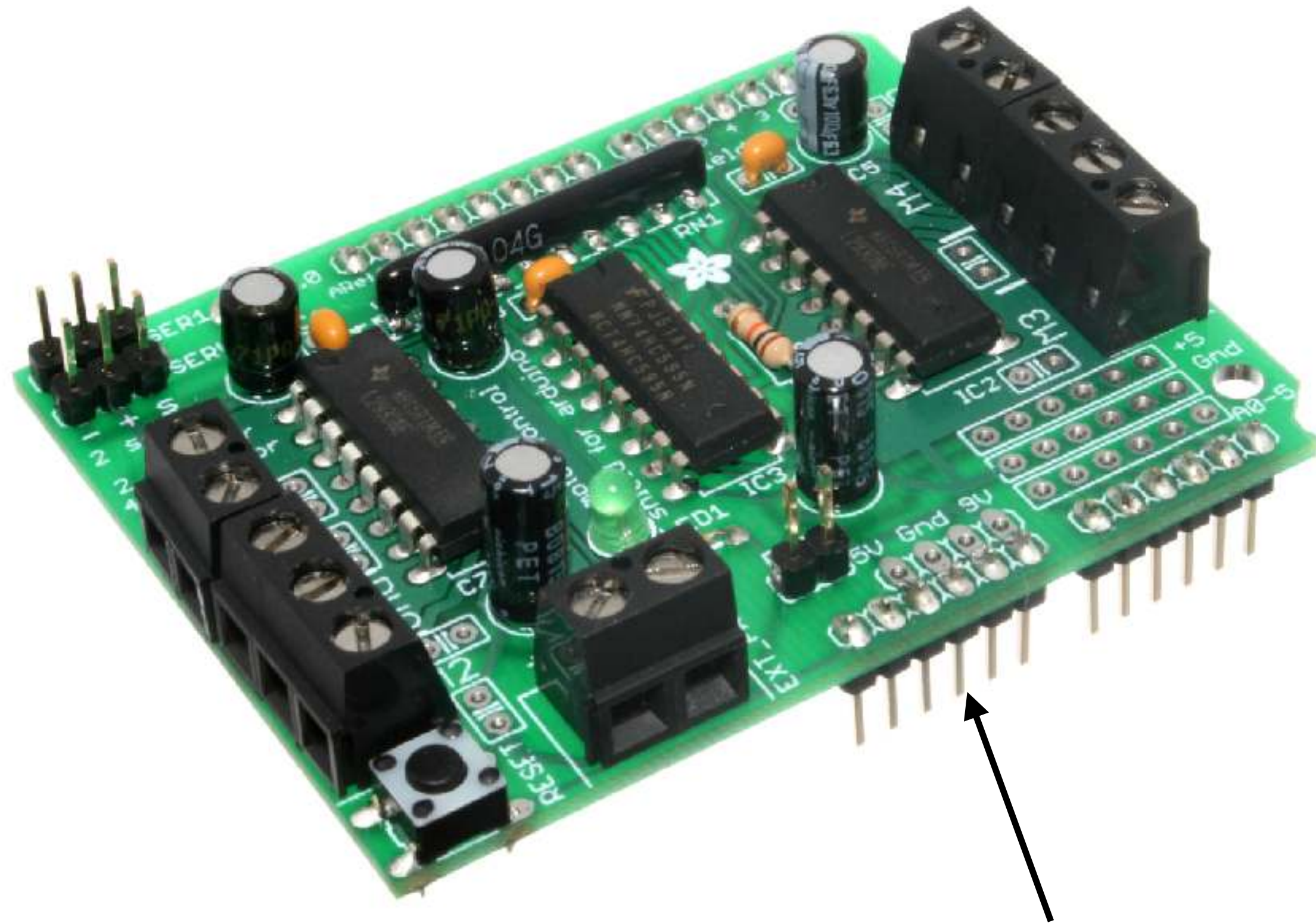


Robot

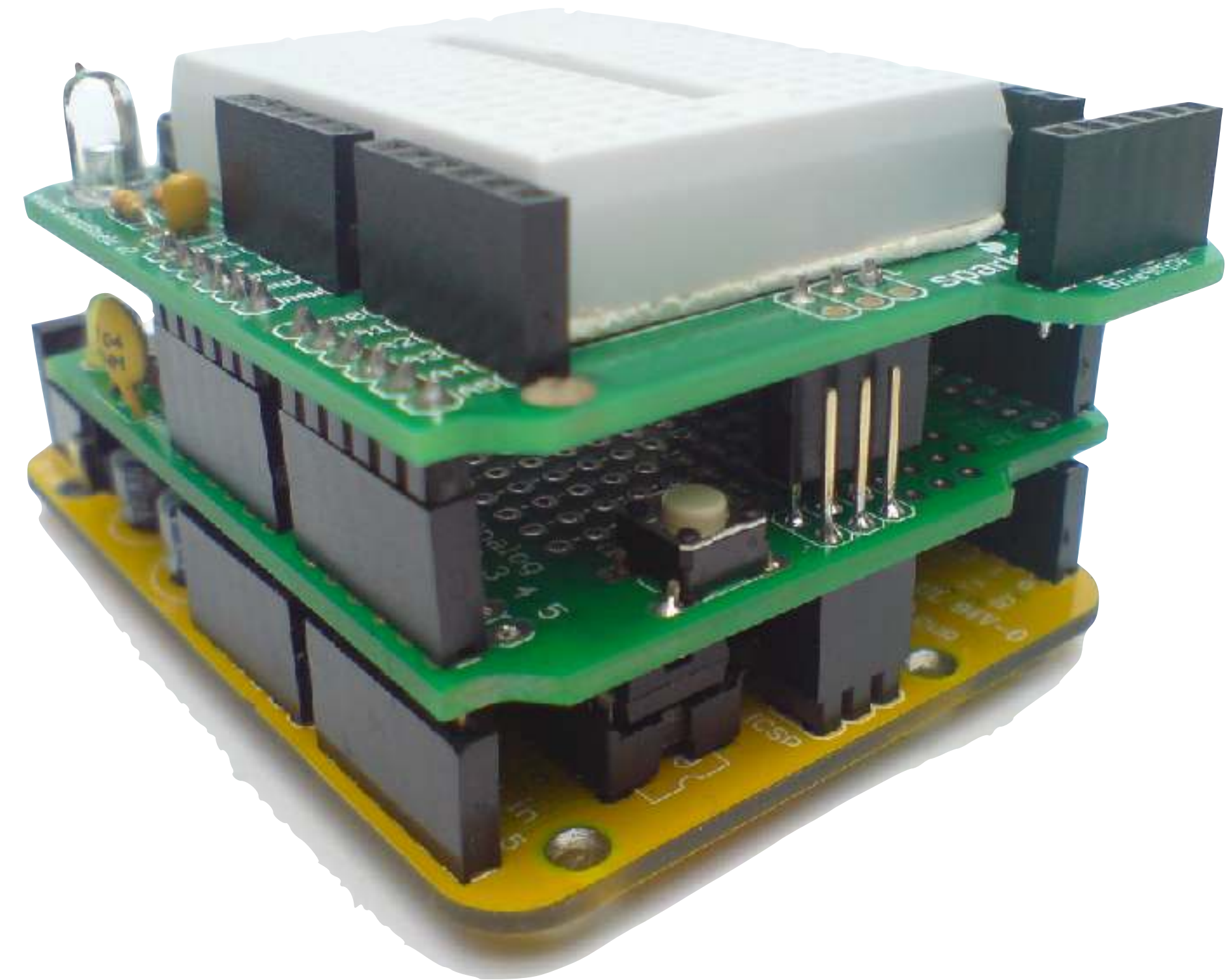
Images: en.wikipedia.org/wiki/Arduino



SHIELDS - EXPAND THE HARDWARE

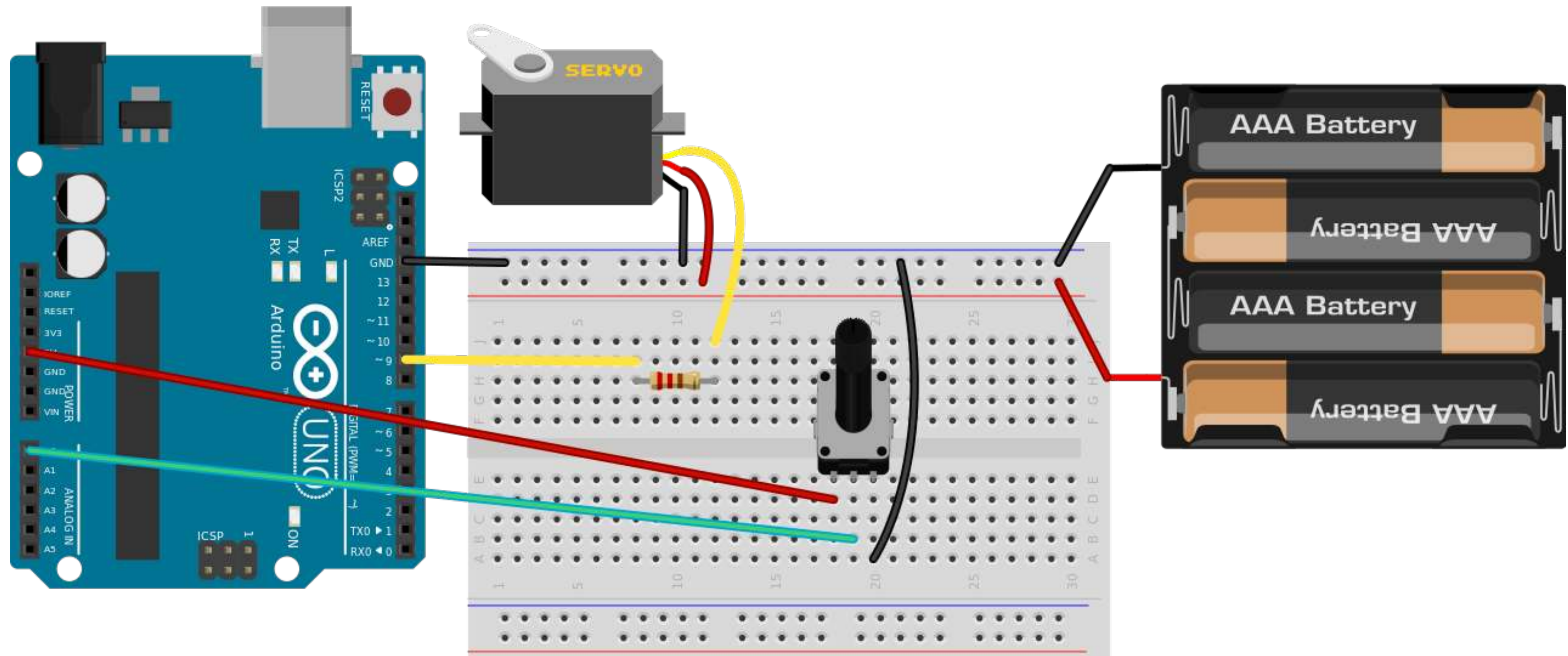


Plugs on top of
micro controller board



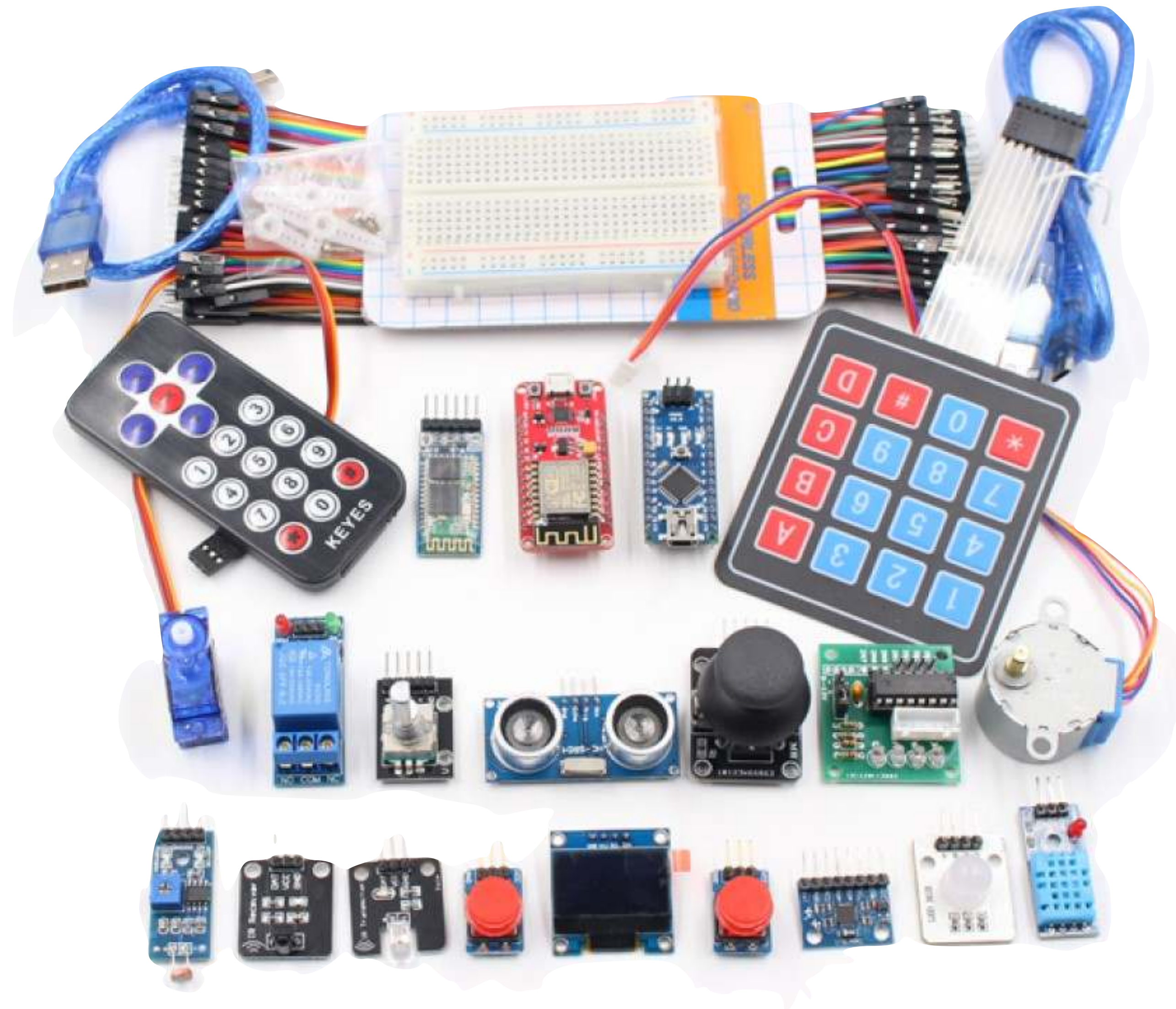
Multiple shields
can be stacked

SHIELDS NOT REQUIRED



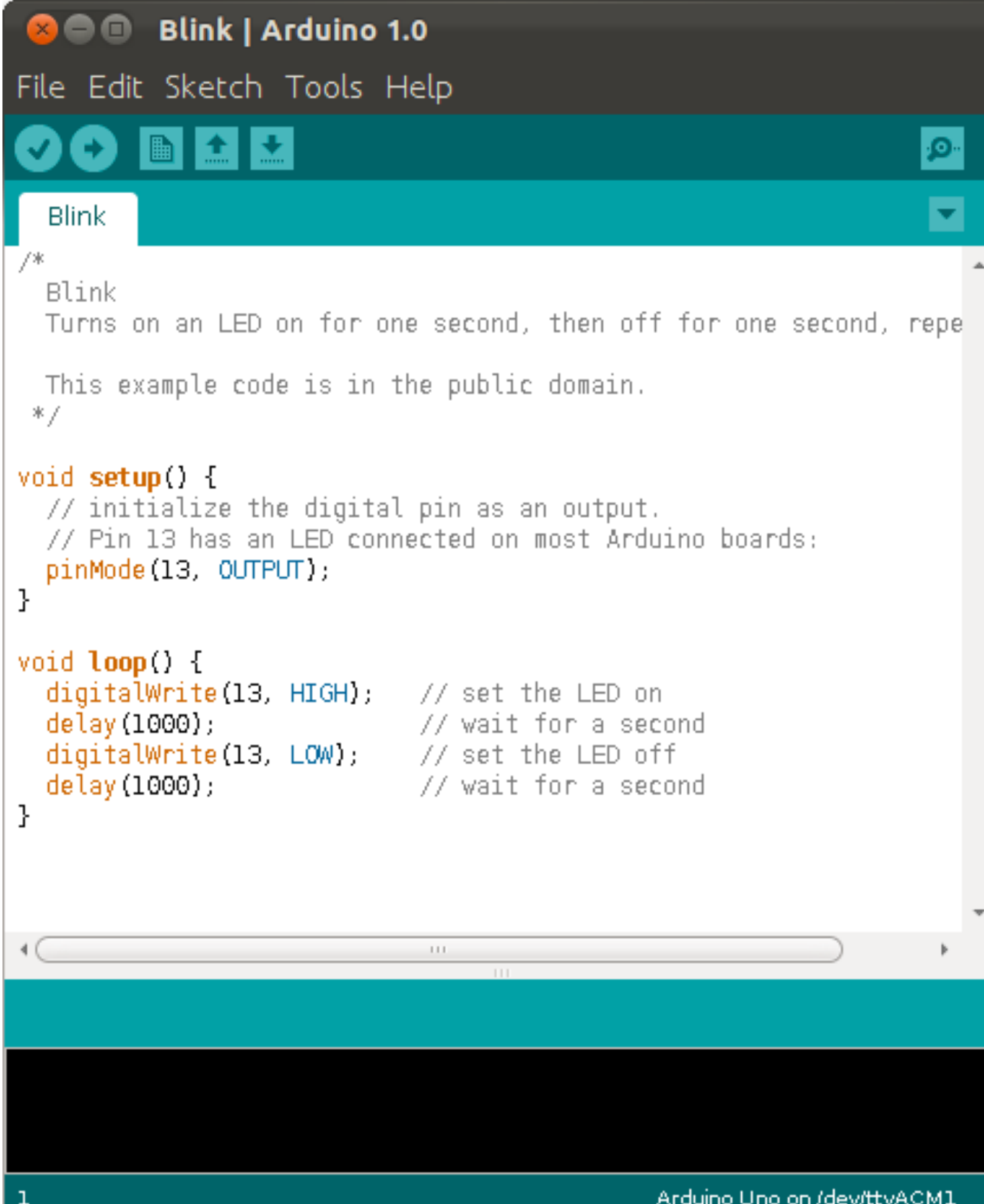
https://commons.wikimedia.org/wiki/File:Servo_motor_Arduino_and_a_potentiometer.svg

TYPICAL COLLECTION OF COMPONENTS



ARDUINO IDE

- The free & open source Arduino IDE
- Uses C / C++ derived language
- Includes libraries
- Supports downloading support for other chipsets and libraries
- A program is called a *sketch*
- Most programs are pretty simple input and output



The screenshot shows the Arduino IDE window titled "Blink | Arduino 1.0". The menu bar includes "File", "Edit", "Sketch", "Tools", and "Help". Below the menu bar is a toolbar with icons for checking, running, uploading, and downloading. The main editor area displays the "Blink" sketch, which is a C++ program that turns an LED on and off in a repeating cycle. The code is as follows:

```
/*
 * Blink
 * Turns on an LED on for one second, then off for one second, repeating.
 *
 * This example code is in the public domain.
 */

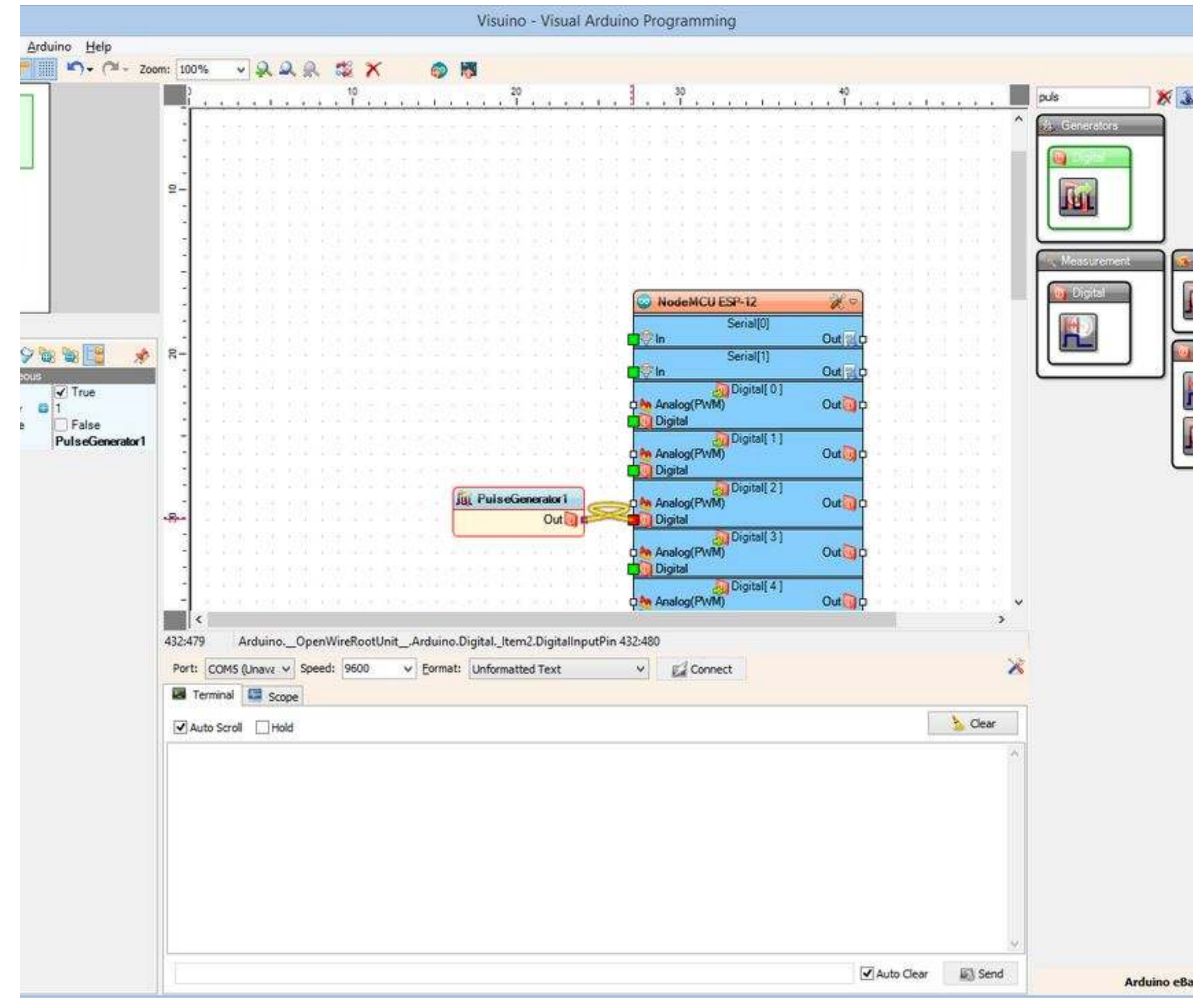
void setup() {
  // initialize the digital pin as an output.
  // Pin 13 has an LED connected on most Arduino boards:
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH); // set the LED on
  delay(1000);            // wait for a second
  digitalWrite(13, LOW);  // set the LED off
  delay(1000);            // wait for a second
}
```

At the bottom of the window, there is a status bar showing "1" on the left and "Arduino Uno on /dev/ttyACM1" on the right.

VISUINO IDE BY MITOV SOFTWARE

- Uses visual design
- Forward engineers to C
- Uses Arduino IDE behind the scenes
- Free to use or \$9 to buy
- Supports most common hardware
- Runs on Windows
- Extensible and expandable
- www.visuino.com





VISUINO DEMOS

Main Site - www.visuino.com

Documentation - www.visuino.com/wiki

Instructables - www.instructables.com/member/BoianM/instructables/

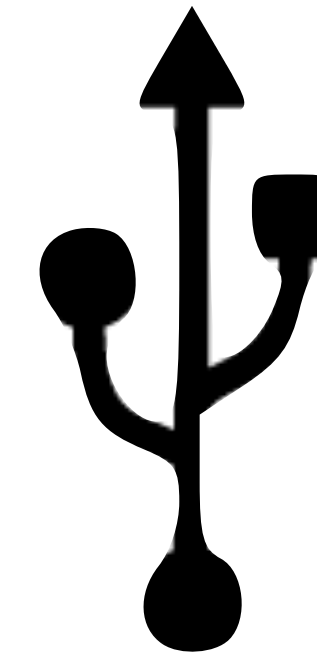
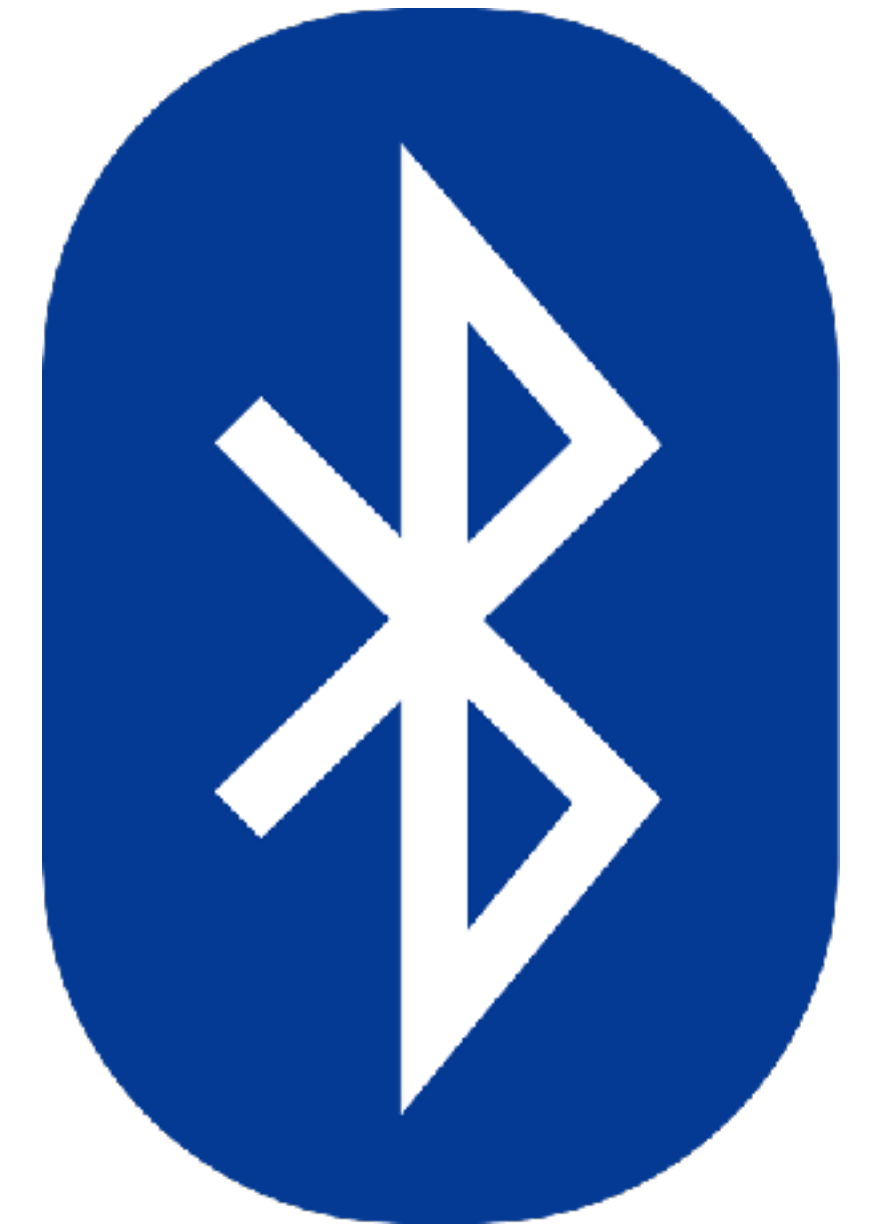
Hackster.IO - www.hackster.io/visuino

YouTube - bit.ly/MitovYT

Blog - labpacks.blogspot.com

ARDUINO COMMUNICATION OPTIONS

- WiFi
- Bluetooth Serial
- Bluetooth LE
- USB Serial
- RS232 Serial
- NFC
- Infrared
- Cellular data
- Other radio

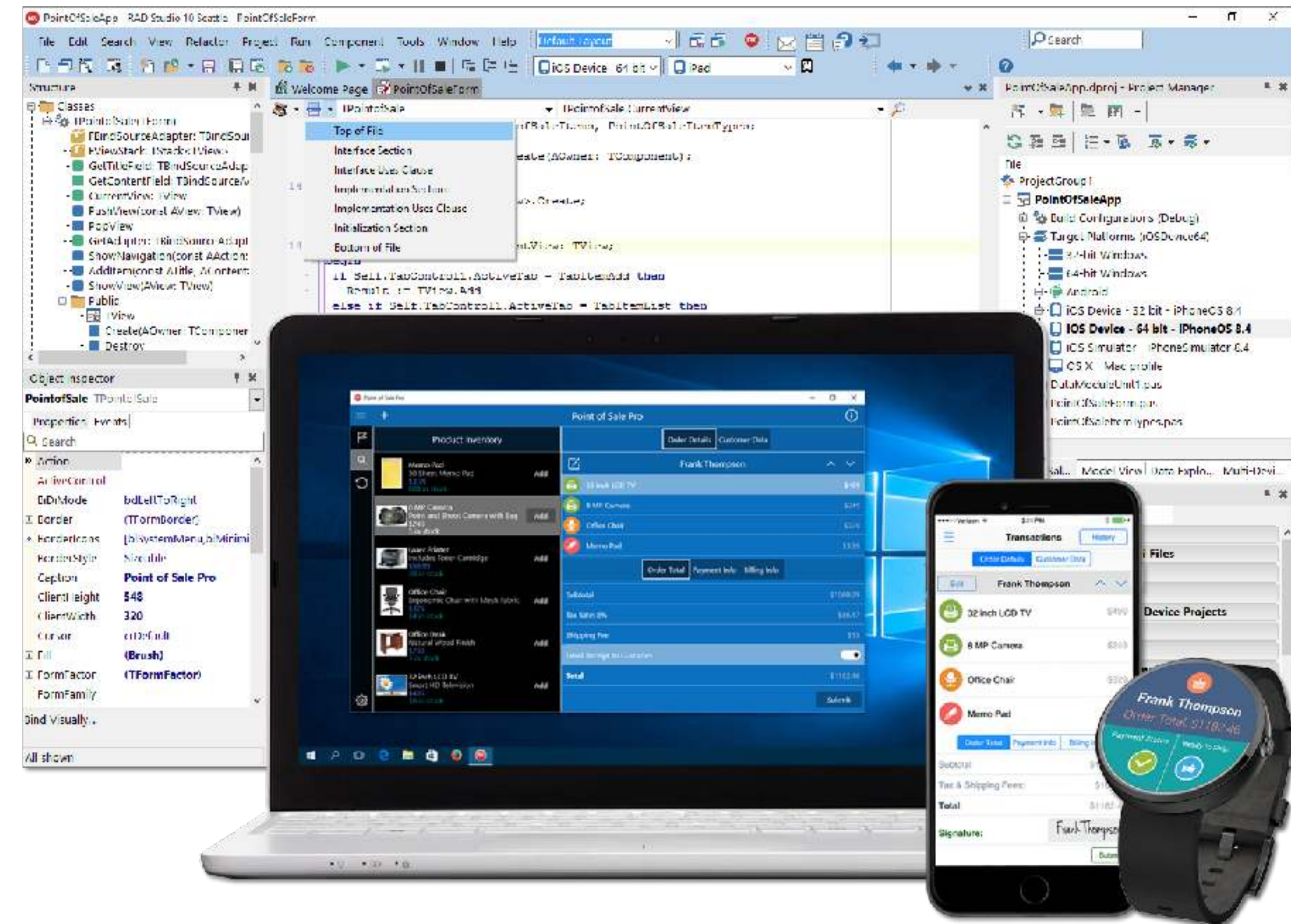


ARCHITECTURAL CONSIDERATIONS

- Direct mobile device communication (Bluetooth or NFC)
- Direct device to cloud (Cellular or WiFi)
- Relay through device to cloud (Bluetooth or WiFi)
- Device to device mesh (other radio)
- Local connection (USB, serial, IR)

RAPID MOBILE APP DEVELOPMENT

- RAD Studio / Delphi / C++Builder
- Supports Windows, macOS, iOS, Android & Linux from same project
- Bluetooth, Bluetooth LE, REST, HTTP, RS232, P2P, etc.
- Rapid visual design
- Native compiled performance
- Included embedded databases
- Free Windows only version
- Paid for other platforms
- www.embarcadero.com



CLOUD OPTIONS

- firebase.google.com 
- [PubNub.com](https://pubnub.com) 
- kinvey.com 
- [ParsePlatform.org](https://parseplatform.org) 
- api.shephertz.com (App42)
- Embarcadero RAD Server (self hosted)
- Amazon, Azure, Google, etc. cloud platforms
- *Roll your own*





APP DEVELOPMENT DEMOS

Using RAD Studio / Delphi

www.embarcadero.com

docwiki.embarcadero.com

community.embarcadero.com

embt.co/IoTBootCamp2017

Q&A

email -> jim.mckeeth@embarcadero.com

[@JimMcKeeth](#) <- Twitter

Blog -> delphi.org

embt.co/loTBootCamp2017 <- Useful link